~~Z Algo~~ ~~Z function~~

Manacher

Tarjan

KMP

Rabin Karp

~~Z function~~

~~Z algo~~

Suffix

Hashing ?

Do hashing last two questions lc

String

Priyansh cp template

1) suffix string → codeforces edu section

Rolling hashing → fla elimination

~~EOL~~ ~~(magartsos)~~
EOF
~~State~~ ~~Yasheh (informatics)~~

stoi(s, iIdx, base)

→ Trim → ~~remove space~~
→ split → stringstream → C++

Cover it

```
vector<string> split (s, d){.
std::stringstream ss(s);
tmp;
while(getline(ss, tmp, d)) vos.pb(tmp);
```

string hashing tle

→ KMP (you should know) Robin Karp
→ Z-Alg] template same.
→ Manacher

KMP

LSP → Longest suffix which is also a prefix

lsp[i] ⇒ LPS for the substring 0 to i.

a	b	a
0	0	1

not considered when we are considering ~~the same~~ a single char

KMP algorithm

Let's use 1 based indexing.

ll lsp[n+1];

lps[i] = s[0 to i]

Longest ~~suffix which is also~~ prefix

a	b	a	c	a	b	a	d
0	0	1	0	1	2	3	0

Do it from algozenith?

KMP algorithm $O(n)$

→ ll i = 0, j = -1;
→ lpsp[0] = -1

→ while (i<n){
$\boxed{\text{while (j != -1 and s[j] != s[i]) j = lpsp[j]}}$
i++;
j++;
lpsp[i]=j;
}

lps array increase not more than one per index change

1) Find occurence of P in T.
- Inbuilt find function in C++
n^2

2) :: Period of a string

→ used in some [illegible] problem

Find P in T.

P = a b a

T = a b a a b c a b a

Soln

S = P + "#" + T

a	b	c	#	a	b	c	a	b	c	a	b	a
				1	2	3	1	2	3	1	2	1

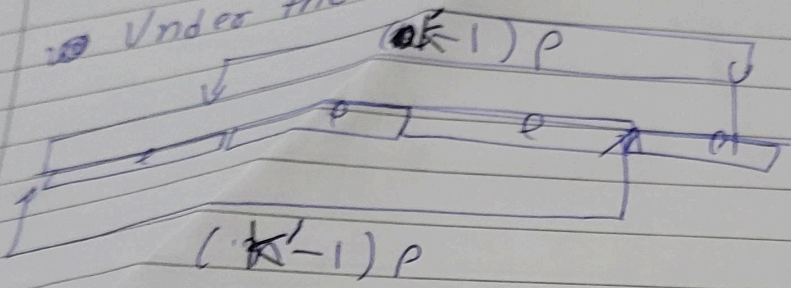
lps

idx == p.size()

idx − p.size() + 1

Justification

Under the hood



it can't match the whole

$m' = (n'-1)p \geq p.$

lps -1
idx 0
n —

Justification for interview

Smallest period of string

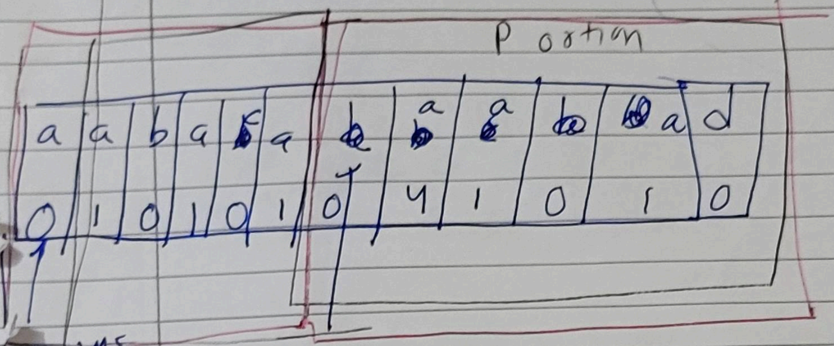
		a	b	c	a	b	c	a	b	c
s	-1	0	0	0	1	2	3	4	5	6
idx	0	1	2	3	4	5	6	7	8	9

$n - lps(n) = \text{period}$

```
if (n % (n - lps(n)) == 0)
n - lps(n)
else
  n
```

Z algorithm ⇒ Z array

Z [i] → largest match b/w



longest len

which matches with prefix
~~(not suffix ending at i)~~

s z

c d e

O(n)

Get readied with a~~n~~ template

Pattern matching (We get starting indices with z algorithm)

P, T

P # T

a b a # a a b a b a c
0 0 1 0 1 3 0 3 0 1 0

z value ∃ len of P

O(n)

Get ...

Q. Minimum letters to be added in the end of the String such that it become palindrome

fundamental

len − y $\quad$ y

$1 \le a$ palindrome $\le n$

s.len()

⇒ Longest suffix which is a palindrome

Q.

s = acebec

⇓

srev + '#' + s

cebeca # $\boxed{acebec}$

lsp(n) is answer

Front of it ?

aace caaa

Biggest palindrome string

S

aacecaaa # aaacecaa

srev.

lps(n)

→ code write for medione

Manachar

arrays

Modd — odd length palindromes with i as middle.

Meven

	a	b	a	c	a	b	a	a	b
Modd(x)	0	1	0	3	0	1	0	0	0
Meven	0	0	0	0	0	0	2	0	

2x+1 gives length of palindrome.

~~2x gives length of palindrome~~

⟹ x give you no. of palindrome with that center

Q1) Longest palindrome in a string.

Q2) No. of palindromes in a string

$\left(\sum M_{odd} + \sum M_{even}\right) + n$

→ as M_{odd} don't count itself